SEAGAS Sprint 3 Summary (for the Sprint 3 / Project A report)

Sprint 3: Integration, Testing and Evaluation · Weeks 11 to 13 (from 2 November 2026) · status on 12 November 2026, after the Sprint 3 meeting with the lecturer. Prepared by Soh Way Miin (Carl) from the code, git history and project documents, so the report can quote checked facts and figures. Owners and statuses in section 3 should be confirmed against ClickUp before they go into the report.

1. How to use this document

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Every figure comes with the file that produced it, so it can be cited or re-run. Where a number is from synthetic students, it says so; the report must not present it as a result with real students.

2. Sprint 3 at a glance

Item	Status on 12 Nov 2026
Sprint goal	Integrate all parts, apply lecturer feedback, test and evaluate, prepare the Alpha build and demonstration.
Meetings	Team rehearsal and final debugging (11 Nov, 3 h); Sprint 3 meeting with the lecturer (12 Nov): full demo with three new students (Cybersecurity, ICT Web and Mobile, Business Marketing).
System scope	3 portals (student, advisor, admin); 3 courses (Computer Science, ICT, Business); 13 programmes; 10 job roles; 172 skills; 377 learning resources; 350 real job descriptions; 5,012 student accounts (5,000 synthetic).
Commits since 1 Nov (dev, no merges)	42: Carl 20, Yong Hin 17, Ren Hang 4, Shanice 1 (Aaron's work is outside git; add from ClickUp).
Automated tests	331 passed, 1 skipped (tests/, 10 test files); frontend production build passes.
Lecturer feedback 12 Nov	8 points: 5 done, 2 in progress, 1 not started (section 4.2).
Still to deliver	F05, F06, F03 validation, performance testing, SUS / UAD evaluation, Sprint 3 report, peer review, poster, video.

3. Sprint 3 backlog (S3-01 to S3-09) and progress

ID	Deliverable	Status	Evidence / what remains
S3-01	Full system integration	Mostly done	Student, advisor and admin portals use live APIs; F01, F02, F07 done; development tracking, reassessment and completion pop-up integrated.
S3-02	NLP and backend integration	Done, verify	Hybrid extraction runs in the backend for CVs, JDs and assessments (start script loads the NLP model). Regression tests in tests/.

ID	Deliverable	Status	Evidence / what remains
S3-03	End-to-end testing	In progress	Full demo flow rehearsed 11 Nov and shown 12 Nov; F04 done; F03 documented, cross-market check open; F05 partly; SUS / UAT not yet run.
S3-04	Bug fixing and edge cases	Ongoing	Section 7. Role-based access covered by backend tests (403/404 checks per role and course).
S3-05	Level 1 / Level 2 matching evaluation	Done (dataset open)	Gold set and method comparison done (section 6.1). F06 larger dataset still to plan.
S3-06	Performance testing	Not started	Needs measured response times (assessment, extraction).
S3-07	Career Path Recommendation	Built earlier	Career Paths page (FR-15, Week 10): match per role, missing skills, next steps. Verify only; not new work.
S3-08	Final Project A report	Not started	This document, docs/JOB_READINESS_SCORE.md and docs/report_drafts/ are inputs.
S3-09	Alpha build and demonstration	Demo done; poster / video open	Alpha build demonstrated 12 Nov; one-click start fixed for all members; poster and video to do.

4. Lecturer feedback and how it was addressed

4.1 From the Sprint 2 meeting (22 Oct), implemented in Week 11

Feedback	What was done
Advisors, not admins, should manage skills, roles and resources	Advisors manage the library of their own courses; seed rows read-only; audit logged.
Skill gaps by programme; soft skills visible	Skill-gap report filtered by programme and category, with a soft-skills view.
A 0% can mean missing data	Scores show "Not required" or "No data yet" instead of 0%.
Explain how the score is made	"How is this calculated?" under each score; skill counts per score; Job Readiness Score write-up.
Justify the weights	Sensitivity analysis on 5,000 assessments (section 6.2).
Reassessment	"Run again" button; readiness over time per role.
Link to MQF	First mapping in Week 11; replaced by the full MQF / COPPA work on 12 Nov (F08).

4.2 From the Sprint 3 meeting (12 Nov)

ID	Feedback	Status	What was done
F01	Filter students by Year 1, Year 2 ...	Done	Year filter and sortable Year column (Advisor → Students).
F02	Advisor sees each student's readiness	Done	Score with assessed role, status and reason per student; dashboard cards open the students behind each number.
F03	Readiness usable internationally	In progress	Formula country-neutral; levels are settings; skills linked to MQF clusters, MQF referenced to AQRF. Open: JDs from more markets and a cross-market check.
F04	Required skills matter more than preferred / bonus	Done	Backend orders by importance (Yong Hin); page grouped Required → Preferred → Bonus with 70/25/5 weights and a reminder.
F05	More detailed recommendations	Partly	Importance, gap type, "up to +X points". Open: why the skill matters for the role; current → expected level.
F06	Larger dataset	Not started	To plan with Yong Hin; current size in section 2.

ID	Feedback	Status	What was done
F07	INTI and Swinburne logos near SEAGAS	Done	Seagas × Swinburne × INTI on home and sign-in; INTI × SWINBURNE × COS40005 in portal sidebars.
F08	MQA referencing and MQA COPPA	Done (first version)	MQF cluster drill-down for advisors; docs/JOB_READINESS_SCORE.md section 4 (MQF, AQRF, COPPA).

5. What was built in Sprint 3

5.1 Assessment, score and recommendations

- **Tiered Job Readiness Score**: required 70% / preferred 25% / bonus 5% of coverage; Developing earns half (Yong Hin; backend/readiness.py).
- **Role templates retuned** against 350 real JDs with hybrid extraction, skills library v4 (Yong Hin; docs/report_drafts/role_template_retune.md).
- **JD fit**: how many real JDs of the role a student already covers, shown beside readiness (Yong Hin).
- **Recommendations ordered by importance** with what each step is worth; grouped Required → Preferred → Bonus with a reminder (Yong Hin, Carl).
- **Completing an activity** needs evidence; a pop-up confirms the skill added, or lets the student raise an existing skill's level; completed skills listed separately (Carl).

5.2 Advisor portal

- **Who needs support**: status compares readiness with the level expected for the student's year and recent activity (On Track, Developing, Getting Started, Improving, Needs Support, Not Assessed), each with its reason; levels are System Settings (Carl).
- **Drill-downs**: every dashboard card opens the students or activities it counts; activity lists grouped by student; MQF cluster → skill → students → student record (Carl).
- **New-student process**: advisors upload the list with student ID and school email, check it, create accounts and send login emails; a duplicate ID or email refuses the whole file; admins view only (Carl).
- Year filter, readiness with role, activity search, aligned skill grids, printable student report (Carl).

5.3 Interface and branding

- Redesigned student interface: student theme and restyled Dashboard, Profile, Job Analysis, Skill Gap, Recommendations, Progress, Career Paths, CV Export (Shanice, PR #30).
- Role-based themes for advisor and admin portals, consistent skill-group colours, sign-in layout, home page styles (Ren Hang, feature/UIUX).
- Brand lockup with Swinburne and INTI logos, sidebar branding and icons, floating-label sign-in, campus background, animated home sections, new font (Carl).

5.4 Platform, data and documentation

- Real email through Gmail SMTP (STARTTLS); tests isolated from the real mail server (Carl).
- Dead learning-resource links replaced and a link checker added (Yong Hin, PR #29).
- One-click start for every member, including folders with non-English names; packages installed on start (Carl).
- Documents: Job Readiness Score write-up (score, sensitivity, status rules, MQF / AQRF / COPPA), System Guide, team updates (Carl); NLP evaluation, method comparison, dataset description, retune report (Yong Hin).

6. Testing and evaluation results (with sources)

6.1 Skill extraction (NLP), Yong Hin

Measure	Result	Source
Gold set	60 Malaysian JDs (10 per IT role), two annotators on 12 JDs	docs/report_drafts/nlp_evaluation.md
Inter-annotator agreement	Positive specific agreement F1 0.867; Cohen's κ 0.763	same
Hybrid, test split (42 JDs, 569 gold skills)	Micro P 0.853, R 0.652, F1 0.739 [0.70, 0.78]	same, Table 1
Keyword, same test	Micro P 0.915, R 0.603, F1 0.727 [0.69, 0.76]	same
Hybrid vs keyword	Not significantly better on F1 (report carefully); hybrid ahead on soft skills (0.743 vs 0.667)	same
Catalogue skills only	Keyword / hybrid F1 0.81–0.82 (recall capped at 0.82 by skills outside the library)	same

6.2 Job Readiness Score (synthetic students)

Measure	Result	Source
Sensitivity of weights (5,000 assessments)	Spearman ρ ≥ 0.987 between tier schemes: the order of students hardly changes	docs/JOB_READINESS_SCORE.md 2.4
Band targets	85.3% of strong archetypes On Track (target ≥ 80%); 76.9% of weak Needs Support (target ≥ 75%), old single bands	same, 2.6
Role retune effect	Mean readiness 50.2 → 45.7 on the dashboard	nlp/experiments/results/role_retune_effect.md
JD fit vs readiness	Spearman ρ 0.826	docs/JOB_READINESS_SCORE.md 7
New status rules (5,012 students)	Needs Support 1,221 → 776; Year 1 275 → 27; 76 new Year 1 students Getting Started	same, 2.5
MQF clusters (latest assessments)	Largest gaps: personal and entrepreneurial 65%, communication 63%; no role asks for ethics and professionalism	same, 4.1

6.3 Software testing

- Automated suite: **331 passed, 1 skipped** (backend API incl. role and course access checks, readiness, recommendation engine, JD pipeline, skills library, gold tools, resource links, synthetic data).
- End-to-end walkthrough with three new demo students (intake → first login → CV → assessment → activity with evidence → reassessment), rehearsed 11 Nov, shown 12 Nov.
- Not yet done: performance measurements (S3-06), SUS usability test, UAD student evaluation.

7. Issues found and fixed in Sprint 3 (selection)

Issue	Fix
A test run used the real mail server settings	Tests clear the mail settings before they start
ICT students given roles not offered to ICT (620 students)	Data generator and migration give ICT roles only (Yong Hin)
Dead learning-resource links	Links replaced; automatic link checker (Yong Hin)
Student search failed with extra spaces in names	Search word by word
start_all.bat: frontend window closed at once in folders with Chinese names	Scripts started by full path; Vite cache cleared; npm install on start
Readiness chart showed one score for all same-day assessments	Points keyed by assessment order

Issue	Fix
A completed recommendation kept appearing for a skill already held	Completion pop-up with level choice; completed skills separated
Almost every Year 1 student flagged Needs Support	Status by year of study and recent activity
Unreadable white text on Change Password; help text in printed report	Colours fixed; help hidden when printing

8. Design decisions worth explaining in the report

- **Decision support, not decision:** advisors see the reason behind every status and can record agree / disagree with each AI result (human oversight).
- **Evidence before credit:** an activity only counts once evidence is attached; an existing skill is raised only if the student chooses.
- **Status relative to year of study:** a low score is normal for new students; Needs Support is kept for those below their year's level and inactive (COPPA 3.3.5).
- **Weights 70 / 25 / 5:** chosen for meaning (required first), then checked: rankings stable under other weights ($\rho \geq 0.987$).
- **MQF clusters as the bridge:** using the framework MQA already referenced to AQRF, rather than claiming international validity ourselves.
- **Advisor-led accounts:** no self-registration; the university's student ID and school email are the identity; duplicates refuse the file.
- **Privacy (NFR-05):** synthetic data for testing and demos; secrets only in backend/.env.

9. Contribution per member in Sprint 3 (from git; confirm with ClickUp)

Member	Main Sprint 3 work
Ng Yong Hin	Tiered readiness score; role template retune (350 JDs, library v4); gold-set evaluation of four methods; JD fit; recommendation ordering; ICT role data fix; resource links and checker; v4 resources; NLP documentation.
Tee Ren Hang	Role-based themes for advisor and admin; skill-group colours; sign-in and home page styling; frontend dependencies.
Yan Min Xuan Shanice	Redesigned student interface (PR #30); team leadership and coordination.
Tan Jun Xiong (Aaron)	No commits in this period's history: add Sprint 3 tasks from ClickUp (e.g. report, recommendation logic F04/F05, poster).
Soh Way Miin (Carl)	Sprint 2 feedback items; advisor-led intake and email; status rules; MQF / COPPA work; advisor drill-downs; completion flow; branding and UI refinement; start scripts; tests; documentation; merging the team's branches.

10. Limitations, remaining work and plan

Limitations to state in the report

- Score, status and band figures are from synthetic students; no evaluation with real students yet (planned SUS and UAD evaluation).
- Hybrid extraction is not significantly more accurate than keyword matching; recall about 0.65; the gold set covers the six IT roles only.
- Job descriptions come from two markets: 65 Malaysian (IT roles) and 285 US LinkedIn postings (all Business roles).
- The MQF mapping of skills is our own preliminary one, not reviewed by MQA or programme owners; no role covers MQF cluster 5.
- Self-reported skills; evidence is checked by advisors, not automatically.

Remaining Sprint 3 work

Task	Suggested owner
F05 detailed recommendations (why the skill matters, current → expected level)	Aaron, Carl
F06 larger dataset; an ethics and professionalism skill in role templates	Yong Hin, Carl
F03 cross-market check (Malaysian vs US JDs of the IT roles first)	Yong Hin, Aaron
S3-06 performance testing with measured response times	Carl
SUS usability test and UAD student evaluation	All
Sprint 3 report, peer and self-assessment, poster, demo video	All

Key documents in the repository: docs/JOB_READINESS_SCORE.md · docs/report_drafts/ (NLP evaluation, method comparison, dataset description, role retune) · data/jd_dataset/README.md · docs/SEAGAS_System_Guide_By_Waymiin.pdf · docs/SEAGAS_TeamUpdate_12November_By_Waymiin.pdf.